

JEE MAIN REVISION MASTER: P-BLOCK ELEMENTS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 25 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet P Block Elements - Core Application	25 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet P Block Elements - Core Application (Very High Yield — 25 PYQs)

Core Principle: Key problem pattern in Kcet P Block Elements.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which of the following halides cannot be hydrolysed?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "NO₂ gas is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "A metalloid is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

NO₂ gas is

- (A) colourless, neutral
- (B) colourless, acidic
- (C) brown, acidic
- (D) brown neutral

Correct Answer: None

Solution Analysis: NO₂ gas is brown and acidic, so the correct choice is: Option C : brown, acidic NO₂ stands for nitrogen dioxide, which is a reddish-brown gas known for its characteristic pungent smell. It is one of the primary pollutants resulting from the combustion of fossil fuels and is a contributor to atmospheric reactions that produce smog and acid rain. As an acidic gas, NO₂ can dissolve in water to form nitric and nitrous acids, contributing to environmental issues such as acid rain.